

OMS Digital Pandemic Preparedness workshop: Agenda

Venue: Oxford Martin School Lecture Theatre, 34 Broad Street, Oxford, OX1 3BD

Monday 21st September

10:00 Registration and coffee
10:30 Session 1: *Welcome and workshop overview*
12:30 Lunch
13:30 Session 2: *UKHSA-led session*
15:00 Break
15:30 Session 3: *UKHSA-led session*
17:00 End

Tuesday 22nd September

09:00 Session 4: *Global perspectives on potential and challenges for digital tools*
10:30 Break
11:00 Session 5: *Ethical, efficient and responsible decision-making*
12:30 Lunch
13:30 Session 6: *Future requirements and opportunities, motivated by past experiences*
15:00 Break
15:30 Session 7: *Global and pathogenic variation in requirements*
17:00 End

19:30 Dinner

Wednesday 23rd September

09:00 Session 8: *The potential of data, and the challenges of collecting and sharing it*
10:30 Break
11:00 Session 9: *Behavioural and social considerations, and ways to measure impacts*
12:30 Lunch
13:30 Session 10: *What next?*
15:00 End

OMS Digital Pandemic Preparedness workshop: Programme

Monday 21st September

Session 1: *Welcome and workshop overview*

- Welcome and general notices
- Christophe Fraser and the OMS Digital Pandemic Preparedness Team

The OMS Digital Pandemic Preparedness project background. Scope and aims for the three days and beyond.

Lightning talks: a rapid round-up of Programme work so far (each 2 slides, 5 mins max):

- Younjung Kim / Christl Donnelly **Federated learning from privacy-preserving data**
- Jamie Webb **A platform approach to ethics for digital pandemic preparedness**
- Michael Parker **Ethics of Digital Pandemic Preparedness**
- Thomas Rawson **Infectious disease and demography**
- Melinda Mills **Future work and Pand-AI**
- Luca Ferretti **Opportunities for epidemiological insights from app data**

12:30 *Lunch*

Session 2: *UKHSA-led session*

13:30 Leanne Cummings *Director of Products and Services for Government Digital Service*

14:15 Steven Riley *Director General and Chief Data Officer, UK Health Security Agency (UKHSA)*

15:00 *Break*

Session 3: *UKHSA-led session*

15:30 Break-out sessions led by Steven Riley

17:00 *End*

Tuesday 22nd September

Session 4: *Global perspectives on potential and challenges for digital tools*

Chair: Peter Horby

9:00 Randy Marsden *former project lead of Exposure Notifications at Apple*

The Triangle of Trust: Lessons from GAEN for the Next Pandemic

9:30 Jason Bay, *National University of Singapore; TraceTogether (Singapore)*

TraceTogether: Reflections on the future of Digital Contact Tracing Apps

10:00 Maria van Kerkhove *Director, Department of Epidemic and Pandemic Management, WHO Health Emergencies Programme*

Learnings and challenges from COVID-19, mpox, Ebola and more: how can digital public health tools help us to prevent and respond quicker for future events

10:30 *Break*

Session 5: *Ethical, efficient and responsible decision-making*

Chair: Michael Parker

11:00 Devi Sridhar *(online), Global Public Health, University of Edinburgh, UK*

A decision-making tree for policy responses to a pathogen with pandemic potential

11:30 Stephen John *Philosophy of Public Health, University of Cambridge, UK*

Pandemic policymaking, multiple perspectives and unexpected consequences: lessons from school closures

12:00 **Panel discussion:** *where does responsibility lie for a digital public health tool and its surrounding infrastructure? How can ethics be embedded? (Devi Sridhar, Stephen John, Peter Horby)*

12:30 *Lunch*

Session 6: Future requirements and opportunities, motivated by past experiences

Chair: Luca Ferretti

13:30 Wolfgang Emmerich Zuhlke (NHS COVID-19 app), Computer Science, University College London, UK

Providing a Digital Contact Tracing service in the first week of a Pandemic

14:00 Justus Benzler Epidemiologist, Robert Koch Institute, Germany (Corona-Warn-App)

Future Requirements and Opportunities for Universal and Frequent Testing - motivated by the German experience with the (imperfect) integration of lab- and community-based testing into an exposure notification app infrastructure

14:20 Vittoria Colizza Director of the Sorbonne Université Modeling Outbreaks Center; INSERM and Sorbonne Université, Paris, France

From mobility traces to epidemic-relevant behaviour: lessons for pandemic preparedness

14:40 Samir Bhatt Machine Learning and Public Health, University of Copenhagen, Denmark and Statistics and Public Health, Imperial College London, UK

AI in Infectious Disease: An overview

15:00 Break

Session 7: Global and pathogenic variation in requirements

Chair: Christophe Fraser

15:30 Njenga Kariuki (online) – Global Health, Washington State University, USA
(suggested) perspectives for the global South, challenges and opportunities for digital tools for LMICs

16:00 Katherine Littler, Global Health ethics and governance, WHO Geneva, Switzerland
(suggested) international governance: how to prepare digital public health tools to be crisis-ready with international interoperability while respecting the ethical requirements and responsibilities of individual public health agencies and technology platforms

16:30 Panel discussion: variation by country, setting and pathogen – opportunities and challenges. How to build trust and create trustworthy digital tools? (Njenga Kariuki, Katherine Littler, Jason Bay, Justus Benzler)

17:00 End

19:00 - 21:30 Dinner at Oxford University Museum of Natural History, Parks Road, OX1 3PW

Wednesday 23rd September

Session 8: *The potential of data, and the challenges of collecting and sharing it*

Chair: Christl Donnelly

9:00 Simon Cauchemez *Institut Pasteur, France*

(suggested) the potential and implications of using data from digital public health devices for epidemiological inference. Benefits and challenges, particularly when compared to conventional inference approaches.

9:20 Navjot Kalra *Director of CEPI Digital*

(suggested) intelligence for value, practical considerations for sharing data and insights, and how to do that across international borders and during a crisis

9:40 James Wilson *philosophy, ethics and public health policy, University College London, UK*

Public health ethics, policy, and digital public health tools: how much, and what kinds of data should be collected? Exploring the trade-offs between maximising population health, ensuring equity, respecting individual privacy, and maintaining trust.

10:00 **Panel discussion:** *trade-offs in data collection and utility, safe data sharing, and the role of AI for digital public health tools (Simon Cauchemez, Navjot Kalra, James Wilson)*

10:30 *Break*

Session 9: *Behavioural and social considerations, and ways to measure impacts*

Chair: Melinda Mills

11:00 Ben Cowling *(online) Epidemiology and Public Health, Hong Kong University, Hong Kong*

Behavioral responses to public health measures, pandemic fatigue, and risk compensation

11:20 Helen Ward *infectious disease epidemiology, Imperial College London, UK*

The use of digital public health tools and remote testing in pandemic preparedness: lessons from the REACT study

11:40 Ciro Cattuto *ISI Foundation, Italy*

Measuring human social networks with wearable devices: lessons from SocioPatterns and the road ahead

12:00 **Panel discussion:** *how can behavioural and social considerations be dynamically captured and embedded in pandemic response? (Ben Cowling, Helen Ward, Ciro Cattuto)*

12:30 *Lunch*

Session 10: *What next?*

13:30 What next?

Chair: Michelle Kendall

Discussion; round-up and plan towards writing a community-led Blueprint for Digital Pandemic Preparedness. Should there be future annual workshops?

15:00 *End*